

From Words to Meaning: A Hands On Exploration of Natural Language Processing

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Abstract—The increasing use of Natural Language Processing in applications such as chatbots, language translation, and text analysis highlights the importance of understanding how machines process and interpret human language. This paper examines the fundamental concepts of text analysis, including tokenization, text preprocessing, Term Frequency Inverse Document Frequency, embeddings, and context vectors. Three hands on exercises were conducted to explore the practical applications of Natural Language Processing through text summarization, language detection, and personally identifiable information detection. The exercises showed that the artificial intelligence model was able to extract relevant details from text, recognize the language used, and identify sensitive personal information such as names, email addresses, and phone numbers. However, limitations were observed when the model processed longer text and mixed language inputs, resulting in errors, unrelated responses, or an uneven confidence score. These findings demonstrate that Natural Language Processing tools can provide fast and useful results, but their outputs still require human review to ensure accuracy and relevance. Overall, the module provided foundational knowledge that can support future laboratory activities, advanced machine learning studies, and the development of a Natural Language Processing based capstone project.

Keywords—*Natural Language Processing, Text analysis, Tokenization, Term Frequency Inverse Document Frequency, Context Vector, Embedding*

I. INTRODUCTION

Machine Learning is one of the leading innovations in the current date. Chatbots and Artificial Intelligence (AI) have become popular in different age groups. With their huge impact on different sectors, AI has different branches and one of them is called Natural Language Processing (NLP), which enables computers to understand and depict human languages. One of the functions of NLP is the text analysis wherein computers were able to extract meaningful words, structure and insights from unstructured text.

Text analysis is often misunderstood. It is not simply analyzing what the text is all about. In the context of NLP, text analysis has different areas. The first area is called Tokenization, wherein the text is broken down into each word with its distinct value, removing stopwords, words that have no value, and reducing words to its base form. The second area is the Statistical Technique, wherein it uses statistical analysis techniques to analyze and counts how frequent the words are used in the text to help identify the

theme or subject of the text. Lastly is the Semantic language model, wherein it represents text as vectors for similarity and finding its related terms using cosine similarity.

Despite the few given functionality of the NLP, my prior knowledge of this topic is limited. My understanding of NLP is just used to extract data and it is used for Chatbots. However, throughout my journey with analytics. I have realized that NLP is not just used for mainly Chatbot and extraction, there is a lot more than that such as language transition, sentiment analysis, text classification, and text summarization. This activity has taught me deeper from the process of tokenization, statistical techniques that can be used, and semantic language models which enable me to understand the process of how computers were able to comprehend human language.

II. EXPERIENTIAL & PRACTICAL INSIGHTS

The given activity has presented three hands-on exercises to explore how text analytics can be used and experience common application of text analytics techniques.

A. Use of Generative AI model to summarize text

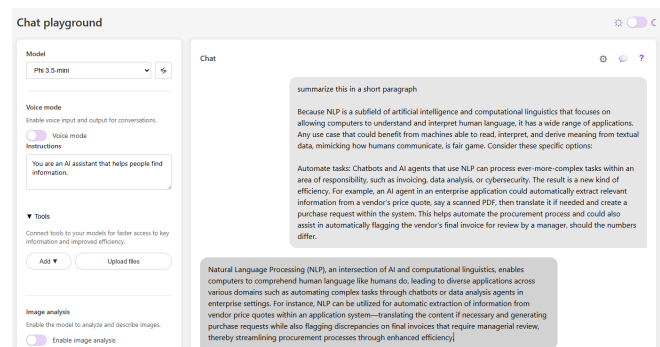


Fig. 1. Exercise 1 Generative AI to summarize text

This exercise is all about creating a concise, summarized text from what I have input. The AI was able to identify the important details that are relevant to the topic and remove other information that is unnecessary. During my experimentation, I tried inputting a longer text, however, I experienced an error while using the model. It also generated an answer that was not related to what my paragraph was about. Through this exercise, I have learned that the quality of the model and length of the input can affect the response generated by the AI. Therefore, it is necessary to read and check what AI produces despite the efficient and quick nature of responses.

B. The use of specialized language analysis tool

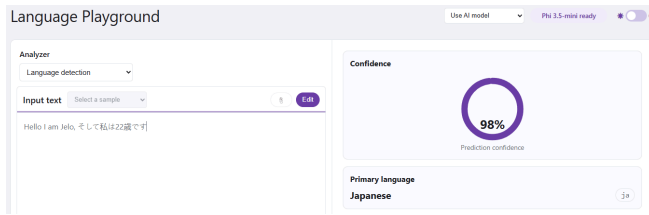


Fig. 2. Exercise 2 Specialized language analysis tool

The second exercise is all about identifying the language of the text. Through my experience, the AI was able to determine and analyze the language I have used. Also, during my experimentation, I tried using different languages and AI was able to predict it correctly. However, I tried using a mixed language, English and Japanese. Although the result was correct, it indicated that the text was 98% Japanese. This result may be because the AI is designed to identify only one primary language or because the Japanese words and characters were more recognizable in the text. Still, the AI was able to identify the dominant language despite the use of two languages.

C. The identification of PII in text

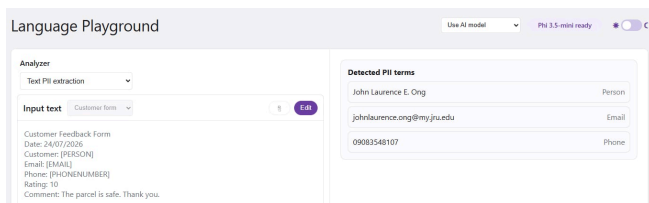


Fig. 3. Exercise 3 Identification of PII in text

The last exercise is about identifying Personally Identified Information or (PII) in text. During my experimentation, I tried using the customer feedback format and input my personal information. The AI was able to identify what I have input. The date, rating and comment input is not included in the detection because it did not contain any relevant identity information. Through this exercise, I was able to understand how PII detection works and how it identifies sensitive details, such as a person's name, email address, and phone number.

Overall, these three exercises helped me understand how AI uses NLP to summarize text, identify languages, and detect personal information. Through my experimentation, I learned that AI can provide useful and accurate results, but it may still produce errors, especially when processing long inputs that the model would not be able to handle properly. A balance intervention from the user and AI would be more efficient in order to produce accurate results.

III. CORE LEARNING OUTCOMES

This module and activity have led me to understand relevant core knowledge that can be useful for future events such as Tokenization, where it breaks text into smaller individual words which are then called tokens. These tokens are then processed into different preprocessing techniques such as normalization, where it removes punctuation and changes all words into smaller cases. Stop word removal, which removes unnecessary words such as “the”, “an” and others which can decrease memory usage. Stemming and Lemmatization which removes end letters such as “ing”, “s”, and “ed” so it will have the same token as the base

form. Despite the clear objective of these techniques, they also have a downside because it can incorrectly divide words, remove useful distinction, remove important meaning, and create invalid words that will affect the performance of the AI. With that, we should select and be careful in using this approach.

Moreover, statistical analysis was also covered in this module. The Term Frequency - Inverse Document Frequency or (TF-IDF), which uses the frequency of words on the text. When two identical documents are presented in order for them to distinguish the difference is by using the TF-IDF. A high TF-IDF means that the word appears frequently only on one document and it is rarely seen in the other document that distinguishes the difference between the documents. On the other hand, the Low TF-IDF score means that the word appears commonly in both documents. Furthermore, the embeddings are used when you want to understand the meaning, context, and the relationships between words. To put it into simpler terms, the TF-IDF uses frequency to distinguish the difference between two things and embeddings are used to distinguish the similarity of two things. Furthermore, a context vector is a more advanced type of embedding, the difference is context vector recognizes that the same word has a different meaning depending on how it was used in the sentence.

IV. COURSE APPLICATION & IMPACT

These given concepts will provide me with the basic knowledge needed for the more advanced use of machine learning models. I believe that the information I have gained will serve as my foundation for future laboratory activities and the capstone project. Understanding tokenization and preprocessing techniques will help me prepare and clean text data before using it in a machine learning model. Moreover, statistical techniques such as TF-IDF will help me identify relevant words for tasks such as text classification. Embeddings and context vectors will also help me analyze the meanings of words and the relationships between them.

Moreover, the exercises in this module have also given me a glimpse of the real applications of NLP that may be considered for future laboratory activities and even the capstone project. These applications include text summarization, text classification, language detection, sentiment analysis, and PII detection. The knowledge and experience I gained from this module will help me prepare data, select the right technique, review the results, and identify possible limitations when developing an NLP-based project. I believe that choosing the right technique is one of the keys to the successful and accurate implementation of machine learning.

V. CONCLUSION & FUTURE WORK

Overall, the hands-on experience from the exercises is the key highlight of this module. Learning how real-world applications work is very intriguing because I have always wondered how platforms such as Google Translate and chatbots are able to understand and accurately respond to the questions or information I provide. With this module, I was partially able to answer that question by learning about tokenization, text preprocessing, statistical techniques, embeddings, and context vectors.

As I was able to understand the basic concepts of NLP, I want to learn how I can build my own NLP model and apply it to real-world problems. As someone considering becoming an analyst in the future, I believe that learning NLP will help me improve my knowledge not only of machine learning but also of deep learning algorithms. I also want to further study Large Language Models such as GPT, which can understand and generate human-like text.